MID-TERM EXAMINATION

(Course Name B.Tech., Semester: - 1st)

(October 2025) OFF LINE mode

Subject Code: BCS 101	Subject: Programming with C
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**	**CO Mappin**
	a) Explain the concept of type conversion and type casting with examples.		**CO1**
	b) Write an expression to find smallest between two numbers using ternary operator.		**CO1**
	c) Write a program to print even numbers from 1 to 15 using a while loop.		**CO2**
	d) What is the purpose of the break statement?		**CO2**
	UNIT I		**CO Mappin**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) Compare relational operators and logical operators with examples.		**CO1**

	b) Analyze the following C expression and explain the order of execution based on operator precedence:
	`int a = 13/3 + 4 * 5 - 7%3;`
	`printf("%d", a);`
			CO1

	c) What are bitwise operators? Explain at least **4 bitwise operators** with examples.		**CO1**
	UNIT II		**CO Mappin**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Write a C program to print sum of all elements of a matrix of size 3×3 using two-dimensional arrays.		**CO2**

	b) Consider the following C program and explain the output:
	`#include <stdio.h>`
	`int main() {`
	`int sum = 0, i;`
	`for (i = 1; i <= 7; i++) {`
	`if (i % 3 == 0) continue;`
	`sum+ = i;`
	`}`
	`printf("Sum = %d", sum);`
	`return 0;`
	`}`
			CO2

	c) Write a C program to find the length of a string without using library function strlen().		**CO2**